12/10/26

First Order Systems $\quad \frac{d\underline{x}}{dt} = \underline{A}\underline{x} \qquad \underline{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ $\quad \underline{A}$ n×n matrix

$\underline{\xi}$ "x_1"

$$\underline{x}(t) = \underline{\xi} e^{rt} \qquad \frac{d\underline{x}}{dt} = \underline{\xi} e^{rt} \cdot r \qquad A\underline{x} = A\underline{\xi} e^{rt}$$

ODE system $\Rightarrow$ $\underline{\xi} e^{rt} r = A \underline{\xi} e^{rt}$

$$\underline{A}\,\underline{\xi} = r\,\underline{\xi} \qquad (4.18)$$

($\underline{\xi}$ → eigenvector)

Eigenvalues : Solve $|\underline{A} - r\underline{I}| = 0$ (eigenvalue) solve for r

Example 4.11 $\quad \frac{d\underline{x}}{dt} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \underline{x}$

$$|\underline{A} - r\underline{I}| = 0 \Rightarrow \begin{vmatrix} 1-r & 1 \\ 4 & 1-r \end{vmatrix} = 0 \Rightarrow (1-r)^2 - 4 = 0$$
$$\Rightarrow 1 + r^2 - 2r - 4 = 0$$
$$(r+1)(r-3) = 0$$

$\therefore$ eigenvalues are $r_1 = -1$ and $r_2 = 3$ //

Eigenvectors

$r_1 = -1$ $\quad (\underline{A} - r_1\underline{I})\underline{\xi}^{(1)} = \underline{0} \Rightarrow \begin{pmatrix} 1-(-1) & 1 \\ 4 & 1-(-1) \end{pmatrix} \begin{pmatrix} \xi_1^{(1)} \\ \xi_2^{(1)} \end{pmatrix} = \underline{0}$

$$\Rightarrow \begin{aligned} 2\xi_1^{(1)} + \xi_2^{(1)} &= 0 \\ 4\xi_1^{(1)} + 2\xi_2^{(1)} &= 0 \end{aligned} \Bigg\} \quad \text{Take } \underline{\xi}^{(1)} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} //$$

[check $A\underline{\xi}^{(1)} = r_1 \underline{\xi}^{(1)}$]

$r_2 = 3$ $\quad (\underline{A} - r_2\underline{I})\underline{\xi}^{(2)} = 0 \Rightarrow \begin{pmatrix} 1-(3) & 1 \\ 4 & 1-3 \end{pmatrix} \begin{pmatrix} \xi_1^{(2)} \\ \xi_2^{(2)} \end{pmatrix} = \underline{0}$

$$\Rightarrow \begin{aligned} -2\xi_1^{(2)} + \xi_2^{(2)} &= 0 \\ 4\xi_1^{(2)} - 2\xi_2^{(2)} &= 0 \end{aligned} \Bigg\} \quad \text{Take } \underline{\xi}^{(2)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} //$$

[Check $A\underline{\xi}^{(2)} = r_2 \underline{\xi}^{(2)}$]

The solution of the differential equation $\frac{d\underline{x}}{dt} = \underline{A}\underline{x}$ are

$\underline{x}^{(1)} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}$ and $\underline{x}^{(2)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t}$ using (4.17)

for all times t. [Check $\frac{d\underline{x}}{dt} = \underline{A}\underline{x}$]

The wronskian

$$W[\underline{x}^{(1)}, \underline{x}^{(2)}] = \begin{vmatrix} \underline{x}^{(1)} & \underline{x}^{(2)} \end{vmatrix} = \begin{vmatrix} e^{-t} & e^{3t} \\ -2e^{-t} & 2e^{3t} \end{vmatrix}$$

$$= 2e^{2t} + 2e^{2t} = 4e^{2t} \neq 0$$

$\therefore \underline{x}^{(1)}$ and $\underline{x}^{(2)}$ are linearly independent solutions (Th^{m} 4.9) and by Th^{m} 4.10 they form a fundamental set and the general solution is

$$\underline{x}(t) = c_1 \underline{x}^{(1)} + c_2 \underline{x}^{(2)}$$

$$\underline{x}(t) = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} \quad \forall t \; //$$

where c_1 and c_2 are arbitrary constants. [Check $\frac{d\underline{x}}{dt} = A\underline{x}$].

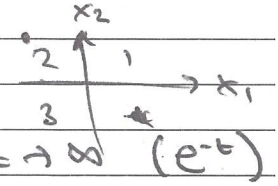
Phase portrait : Graph in the x_1x_2-plane for various c_1 and c_2.

(i) Consider $c_2 = 0$ $\quad \underline{x}(t) = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}$

Component form

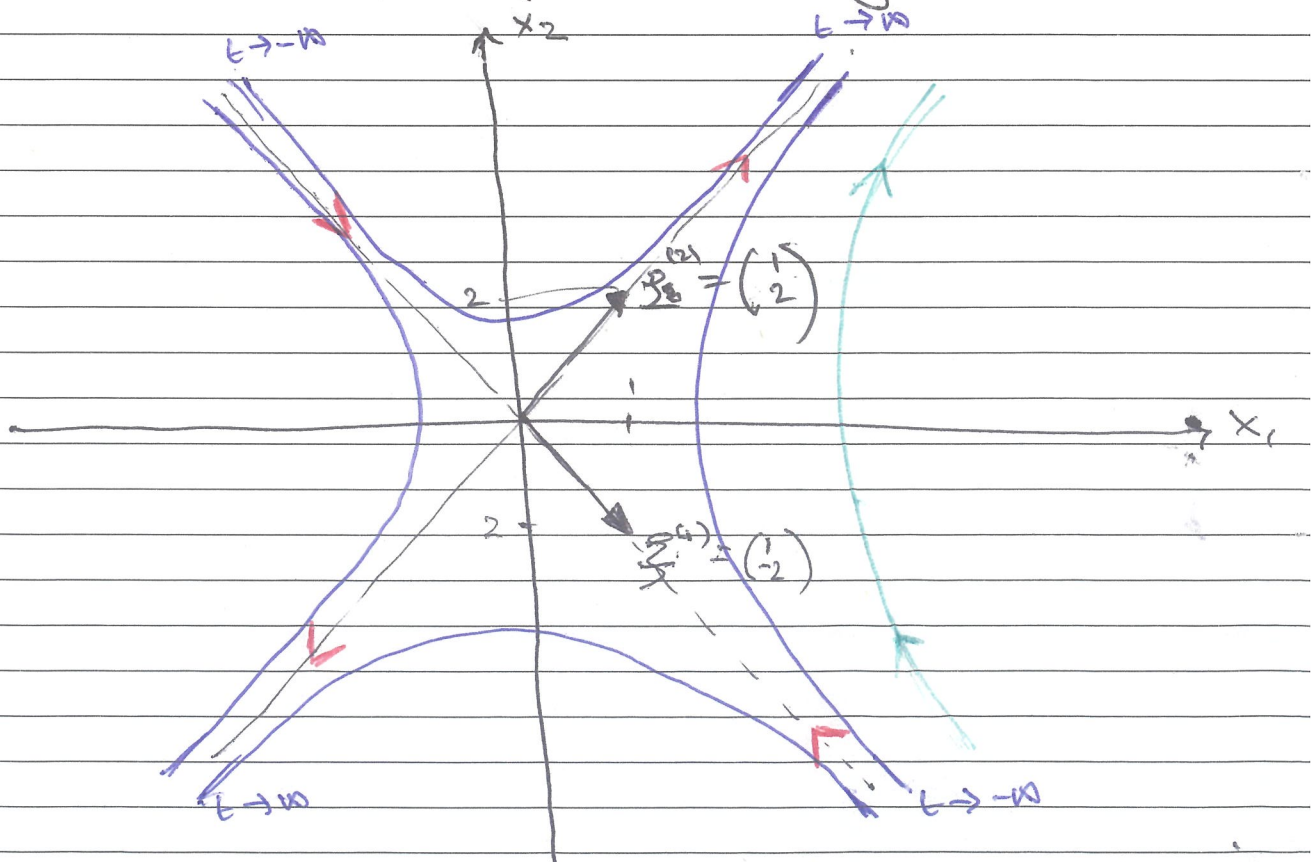
$$x_1 = c_1 e^{-t} \quad \text{and} \quad x_2 = -2c_1 e^{-t}$$

Eliminate t: $\quad x_2 = -2x_1$ straight line through $\underline{0}$ with gradient -2 in the direction of $\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \underline{\xi}^{(1)}$



- $c_1 > 0$ solution is in the 4th quadrant
  $c_1 < 0$ 2nd quadrant

In either case, solution approaches the origin as $t \to \infty$



Real and opposite sign eigenvalues : origin is a saddle point

(ii) $x_1 = 0$ $\underline{x}(t) = c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t}$ $x_1 = c_2 e^{3t}$ $x_2 = 2c_2 e^{3t}$

$x_2 = 2x_1$

straight line parallel to $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$c_2 > 0$ 1st quadrant

$c_2 < 0$ 3rd quadrant

In either case, solution moves away from $\underline{0}$ as $t \to \infty$

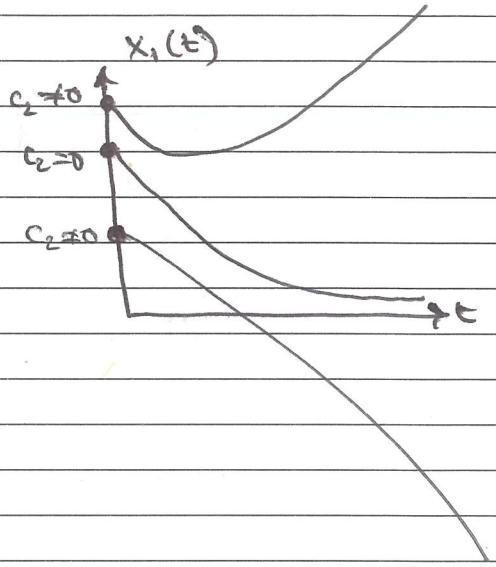
(iii) Combine $\underline{x}(t) = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t}$ General soln

For large t, $\underline{x}^{(2)} >> \underline{x}^{(1)}$ $\therefore$ all solutions are asymptotic to $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ as $t \to \infty$

For $t \to -\infty$, all solutions are parallel to $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$

Plot $x_1(t) = c_1 e^{-t} + c_2 e^{3t}$

- $c_2 = 0$ $x_1(t) \to 0$ as $t \to \infty$
- more generally $c_2 \neq 0$, as $t \to \infty$ $c_2 e^{3t}$ dominates and solution moves away from origin



Plot of $x_2(t)$?